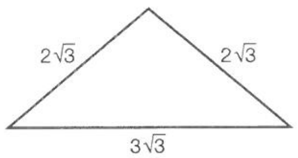
Open-ended – Sample answers

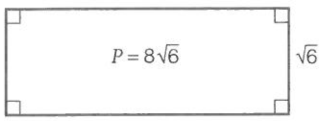
10 $7\sqrt{5} - 4\sqrt{5} = 3\sqrt{5}$; $-10\sqrt{5} + 13\sqrt{5} = 3\sqrt{5}$; $8\sqrt{5} - 5\sqrt{5} = 3\sqrt{5}$

11 length $= 2\sqrt{6}$ cm, width $= 2$ cm, perimeter $= 4\sqrt{6} + 4$ cm;
length $= 2\sqrt{3}$ cm, width $= 2\sqrt{2}$ cm,
perimeter $= 4\sqrt{3} + 4\sqrt{2}$ cm

12 (a) (i) Ruby has added the numbers underneath the square root.
(ii) When subtracting one surd from another, Ruby has forgotten to include the surd part of the solution.
(b) (i) $7\sqrt{3}$ (ii) $3\sqrt{6}$
(c) (i)



(ii)



Exercise 12.4 (p. 678)

1 (a) $3\sqrt{7} + 6$ (b) $-4\sqrt{11} - 4\sqrt{3}$
(c) $18\sqrt{5} - 9\sqrt{2}$ (d) $3\sqrt{35} - 5\sqrt{14}$
(e) $8\sqrt{10} + 24\sqrt{6}$ (f) $3\sqrt{10} - 21\sqrt{15} + 9\sqrt{35}$
(g) $3 + 2\sqrt{3}$ (h) $15\sqrt{14} - 70$
(i) $24\sqrt{15} - 6\sqrt{5} - 15$ (j) $6\sqrt{7} - 12$
(k) $12 - 12\sqrt{2} + 20\sqrt{3}$ (l) $40 - 4\sqrt{5} - 20\sqrt{3}$
(m) $\sqrt{10} + 3\sqrt{5} + 4\sqrt{2} + 12$ (n) $3 + \sqrt{21} - \sqrt{15} - \sqrt{35}$
(o) $5 + 2\sqrt{6}$ (p) $88 - 16\sqrt{10}$
(q) 25 (r) -53

2 (a) B (b) D (c) C (d) C

3 (a) $\pi(19 - 6\sqrt{10})$ m^2 (b) $(2\sqrt{35} + 12\sqrt{7})$ cm^2

4 $\pi(21 + 8\sqrt{5})$ cm^2

5 (a) 21 km^2 (b) $(14 + 3\sqrt{2} + \sqrt{10})$ km
(c) 21 405 m

6 (a) (i) $(18\sqrt{2} - 12)$ m (ii) $(12\sqrt{2} - 8)$ m
(iii) $(15\sqrt{2} - 10)$ m (iv) $(21\sqrt{2} - 14)$ m
(b) (i) 13.46 m (ii) 8.97 m
(iii) 11.21 m (iv) 15.70 m

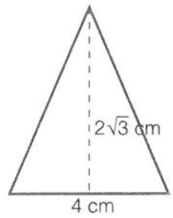
7 (a) 7π cm^2 (b) $(5 + 2\sqrt{7})$ cm by $(3 + 2\sqrt{7})$ cm
(c) $(43 + 16\sqrt{7})$ cm^2 (d) $(43 + 16\sqrt{7} - 7\pi)$ cm^2
(e) 63.34 cm^2

8 (a) 47.7 cm (b) 22 752.9 cm^2
(c) $(12\,110 + 1440\sqrt{55})$ cm^2 (d) 22 789.3 cm^2
(e) part (d)

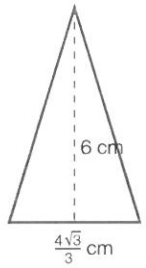
9 (a) $\sqrt{18}$ or $3\sqrt{2}$ cm (b) $150\sqrt{2}$ cm (c) 212.1 cm
(d) 4.2 cm (e) 210 cm (f) part (c)

Open-ended – Sample answers

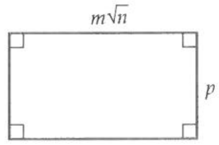
10 base 4 cm, height $2\sqrt{3}$ cm



base $\frac{4\sqrt{3}}{3}$ cm, height 6 cm



11 (a)



$P = 2(m\sqrt{n} + p)$
If $m = 10$, $n = 3$ and $p = 4$
Then $P = 2(10\sqrt{3} + 4) = 42.641$ cm
(b) $A = m\sqrt{n} \times p = mp\sqrt{n}$
(c) If $m = 2$, $n = 2$ and $p = 3$ then $A = 8.485$ m^2

Exercise 12.5 (p. 684)

1 (a) $\frac{\sqrt{7}}{7}$ (b) $\frac{4\sqrt{3}}{3}$ (c) $-2\sqrt{6}$ (d) $\frac{\sqrt{6}}{2}$
(e) $\sqrt{55}$ (f) $\frac{\sqrt{30}}{2}$ (g) $-\frac{3\sqrt{7}}{28}$ (h) $\frac{2\sqrt{15}}{7}$

2 (a) $\frac{3 + \sqrt{7}}{2}$ (b) $8 - 4\sqrt{3}$
(c) $\frac{\sqrt{15} + \sqrt{6}}{3}$ (d) $\frac{4\sqrt{7} - 10}{3}$
(e) $\frac{24 + 15\sqrt{3}}{3} = 8 + 5\sqrt{3}$ (f) $\frac{22 + 10\sqrt{5}}{4} = \frac{11 + 5\sqrt{5}}{2}$
(g) $\frac{6\sqrt{3} - \sqrt{42} - 3\sqrt{2} + \sqrt{7}}{11}$ (h) $\frac{5\sqrt{6} - 8\sqrt{3}}{6}$

3 (a) D (b) A (c) B

4 (a) B (b) D

5 (a) $\frac{2\sqrt{3}+\sqrt{5}+1}{4}$ (b) $\frac{\sqrt{7}+2\sqrt{6}+5}{6}$

(c) $\frac{5\sqrt{6}-5\sqrt{3}-12-3\sqrt{11}}{15}$ (d) $\frac{3\sqrt{5}-3\sqrt{3}-2\sqrt{7}-4}{6}$

6 (a) $\frac{5\sqrt{3}}{2}$ (b) 4 shelves (c) $(5\sqrt{6}-8\sqrt{2})$ m

7 (a) $15\,000\sqrt{6}$ cm^2 (b) $50\sqrt{6}$ tiles

(c) As $122 < 50\sqrt{6} < 123$, the minimum number of tiles is 123, but maybe buy 130 tiles to allow for wastage.

(d) $15\,000\sqrt{6}+6000\sqrt{15}+600\sqrt{35}$ cm^2

8 $\frac{112\sqrt{3}-294}{249}$

Open-ended – Sample answers

9 Many answers are possible, just multiply the surd by a form of 1. As examples: $\frac{4\sqrt{6}}{3}, \frac{8}{\sqrt{6}}, \frac{4\sqrt{10}}{\sqrt{15}}$

10 (a) $\frac{\sqrt{2}}{3}, \frac{2}{3}$ (b) $\frac{2\sqrt{7}}{7}, \frac{4\sqrt{7}}{7}$

(c) $\frac{\sqrt{15}}{5}, \frac{3\sqrt{15}}{5}$ (d) $\frac{\sqrt{2}}{6}, \frac{\sqrt{2}}{2}$

11 $2-\sqrt{3}$ and $4-2\sqrt{3}$

12 When two surds that are the same are multiplied, the result is a whole number, e.g. $\sqrt{6}\times\sqrt{6}=6$. Multiplying by the conjugate will remove the square roots from the denominator, resulting in a rationalised denominator. It is also like the difference of two squares.

Half-time 12 (p 688)

1 (a) 6 (b) 88 (c) $\frac{9}{4}$

2 $5\sqrt{6}+3\sqrt{2}$ m

3 (a) $\frac{\sqrt{30}}{6}$ (b) $\frac{5\sqrt{3}}{6}$ (c) $\frac{12+3\sqrt{11}}{5}$

4 $3\sqrt{5}$ m

5 (a) rational (b) rational (c) irrational (d) rational

6 $(18+12\sqrt{15})$ m^2

7 (a) $6\sqrt{2}$ (b) 360 (c) $\frac{3}{5}$

8 (a) -15 (b) 17 (c) 34

Exercise 12.6 (p. 690)

1 (a) $12x^5$ (b) $\frac{a^5}{b^3}$ (c) $\frac{p}{q^3}$

(d) x^2 (e) $a^{20}b^9$ (f) $6m^{10}n$

(g) $\frac{1}{q^7}$ (h) $\frac{1}{c^9d^5}$ (i) 2

(j) $\frac{a^2}{b^2}$ (k) $\frac{m^5}{n^5}$ (l) 1

(m) q (n) $\frac{3k^3}{10}$ (o) $\frac{2a^6b^4}{3}$

2 (a) k^7p^6 (b) x^5y^5 (c) $\frac{6m^6n^5}{5}$

(d) $\frac{2q^{11}}{p^4}$ (e) $\frac{7y^{10}}{x^4}$ (f) $\frac{1}{m^6n^2}$

(g) $\frac{1}{k^7p^4}$ (h) $\frac{2a^5}{3b}$ (i) $\frac{5fg^4}{2}$

(j) x^2y^4 (k) $\frac{2m^3}{3}$ (l) $\frac{q^2}{5}$

3 (a) B (b) A (c) C

4 (a) x^9y^4 (b) $\frac{a^{12}}{b^{11}}$ (c) $\frac{4}{k^{18}p^2}$

(d) $9m^{20}n^2$ (e) $\frac{x^{27}}{y^{11}}$ (f) $\frac{m^6}{n^{20}}$

(g) $\frac{1}{q^{22}}$ (h) f^6g^{11} (i) $\frac{a}{b^7}$

(j) $\frac{m^{14}}{n^{11}}$ (k) $\frac{4x^{13}}{y^{18}}$ (l) $\frac{12d^{15}}{c^3}$

5 (a) x^2 (b) $\frac{a^3}{b^5}$ (c) pq (d) d^4f^2

(e) $\frac{1}{y^2}$ (f) $\frac{1}{a^3}$ (g) $\frac{n}{m^2}$ (h) $\frac{q^2}{p^7}$

(i) $\frac{b^9}{a^{16}}$ (j) $\frac{x^{13}}{y^7}$ (k) $\frac{12q^4}{p^{17}}$ (l) $\frac{9y^7}{8x^{14}}$

6 (a) $5^3, 5^2, 5^1, 5^0$ (b) $5^{-1}, 5^{-2}, 5^{-3}, 5^{-4}, 5^{-5}$

(c) $125, 25, 5, 1, \frac{1}{5}, \frac{1}{25}, \frac{1}{125}, \frac{1}{625}, \frac{1}{3125}$

7 (a) $\frac{m^9}{n^2}$ (b) $\frac{b^3}{a}$ (c) $\frac{x^5}{y^4}$ (d) $\frac{y^{19}}{x^6}$ (e) $\frac{9}{a^2b^7}$

(f) $\frac{8}{d^5f^7}$ (g) $\frac{10b}{a^9}$ (h) $\frac{n^3}{m^8}$ (i) $\frac{1}{x^2}$

8 (a) 1 divided by the original number (b) $192a$

Open-ended – Sample answers

9 (a) $(6m^2n^3)^2\times n^2$; $9m^3n^4\times 4mn^4$

(b) $\frac{12m^7n^3}{15m^3n^9}$; $24m^4\div 30n^6$

10 $a=2, b=3$; $a=3, b=2$; $a=5, b=1$

These give the fractions: $\frac{1}{8}, \frac{1}{9}$ and $\frac{1}{5}$